Optimization and Curve Sketching

Critical points and the first/second derivative tests

1 Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

Optimization applications

3 A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

4 An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.